AI-Based Medical Image Diagnosis

A computer-aided medical image analysis system using Computer Vision and Deep Learning

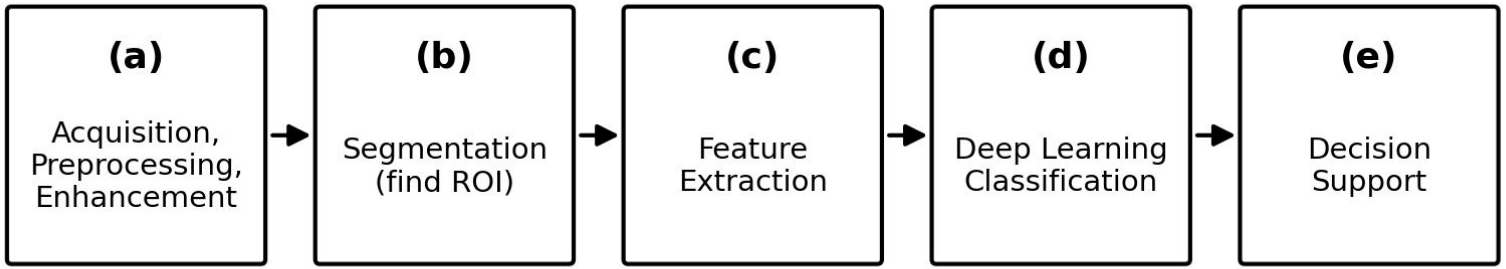
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Introduction

- Doctors use X-ray, CT, MRI and ultrasound images to find diseases
- Reading thousands of images by eye is slow, tiring and can vary from person to person
- Computer vision lets a computer see and measure what is inside an image
- Deep learning lets it learn disease patterns from thousands of examples
- Aim: build a system that helps doctors read images faster and more accurately

Overall System Pipeline

Five stages from raw scan to doctor's report



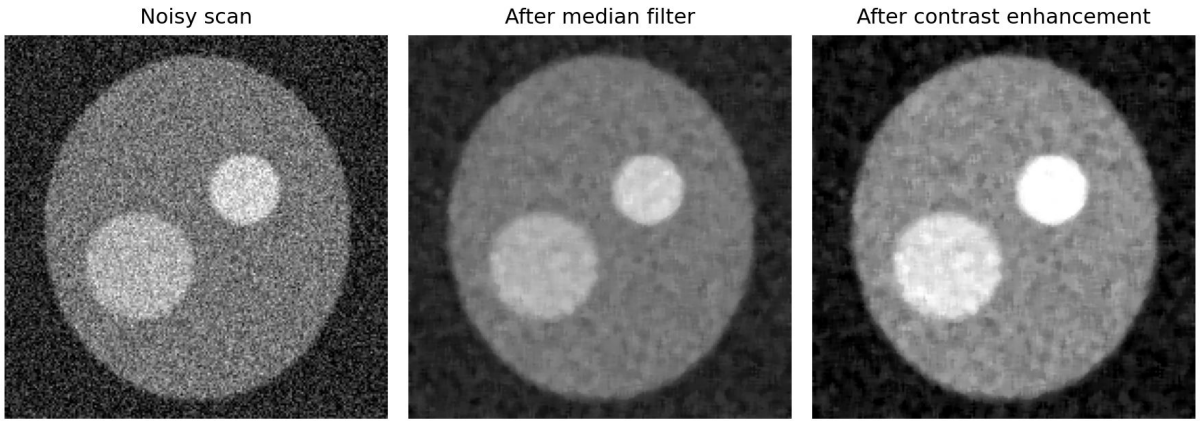
- Scan goes in, cleaned and enhanced image comes out of stage (a)
- Stages (b) and (c) find the problem area and describe it with numbers
- Stage (d) decides what the pattern means; stage (e) reports it to the doctor

(a) Image Acquisition and Preprocessing

- First, we collect the medical images — like X-rays, CT scans, MRI, or ultrasound.
- We store the images in DICOM format and remove patient details to keep their information private.
- Next, we make all images the same size so the neural network can process them easily.
- We adjust the brightness so images from different scanning machines look more similar.
- Finally, we flip, rotate, or crop some images to create more training examples for the AI.

(b) Noise Removal and Enhancement

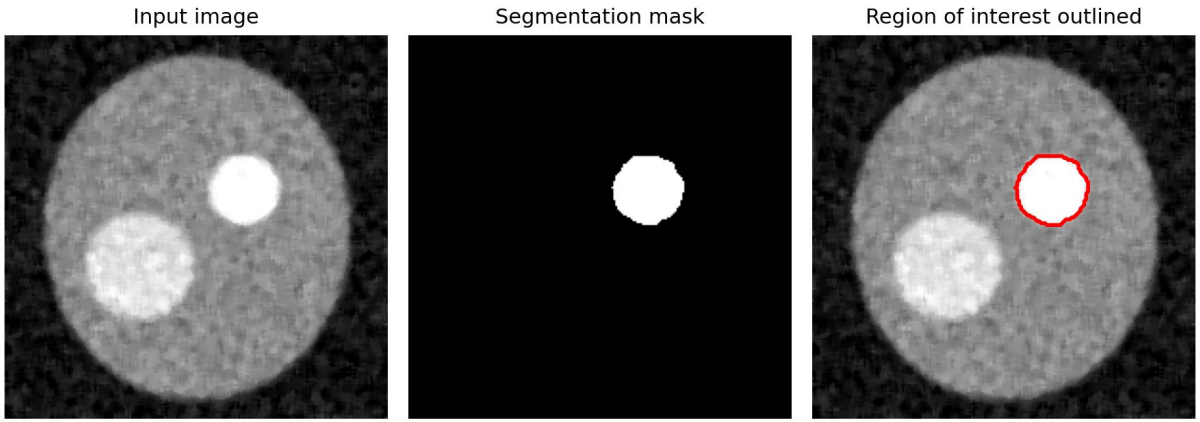
Example on a synthetic scan



- Medical images may contain unwanted noise due to the scanning process. Different scans can have different types of noise.
- We use filters like Median, Gaussian, and Bilateral to remove unwanted noise from the images.
- Finally, we improve the image contrast using Histogram Equalisation or CLAHE, making important details and edges easier to see.

(c) Segmentation of Regions of Interest

Example on a synthetic scan



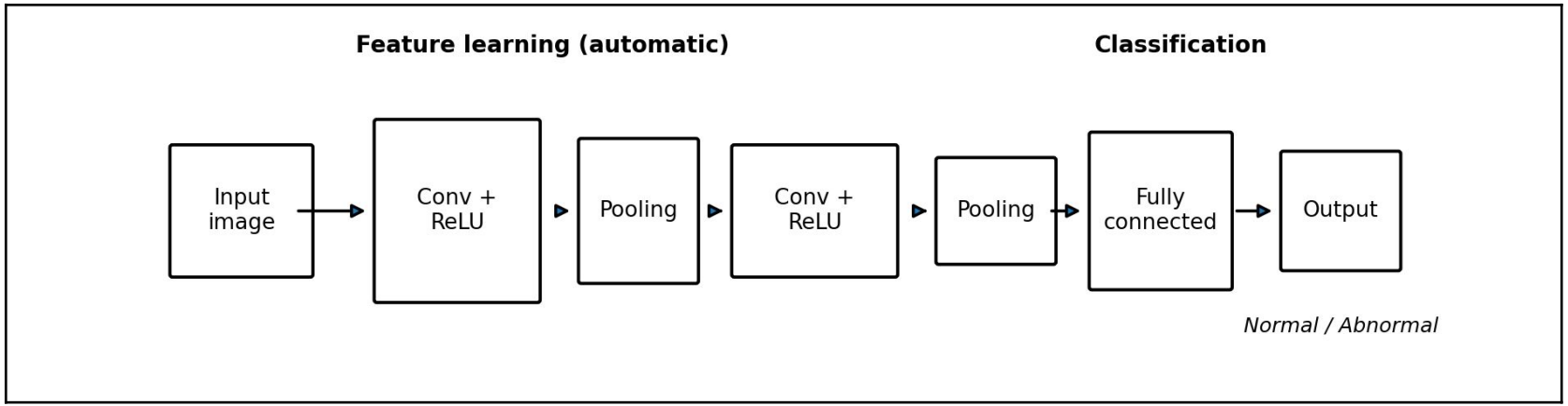
- Segmentation helps separate important areas, such as tumours, lesions, or organs, from the background.
- We can use traditional methods like thresholding and region growing, or deep learning models like U-Net, SegNet, and Mask R-CNN to identify these areas.
- We measure segmentation accuracy using Dice score and IoU to check how accurately the system identifies the required region.

(d) Feature Detection and Extraction

- Converts the selected part of the image into useful information that the computer can understand.
- **Shape features:** Looks at the shape and size, such as the area, boundary, and how regular or irregular it is.
- **Texture features:** Checks the surface pattern of the tissue, such as whether it looks smooth or rough.
- **Intensity features:** Looks at the brightness of the pixels, including their average and how much they vary.
- **Deep learning:** Instead of manually selecting features, deep learning models can automatically learn important features from the image.

(e) Deep-Learning Classification

Convolutional Neural Network (CNN) layout



- CNN learns patterns from medical images, starting with simple edges and then identifying shapes related to diseases.
- We use models like ResNet, DenseNet, VGG, and EfficientNet. Transfer learning helps when we have fewer training images.
- Finally, the system predicts whether the image is normal or abnormal, identifies the possible disease type, and provides a probability score.

(f) Automated Analysis and Decision Support

- The system generates a report showing the affected area, its measurements, and a confidence score.
- Grad-CAM heat maps highlight the parts of the image that influenced the AI's decision.
- If a case looks urgent, the system flags it so doctors can review it first.
- The system can connect with hospital systems like PACS to display the results alongside the scan.
- The doctor makes the final decision. AI only helps the doctor by providing a second opinion.

Benefits and Challenges

Benefits	Challenges
• Faster diagnosis • Consistent results • Detects small, early signs • Reduces doctor workload	• Limited labelled medical data • Different scanners give different images • Model decisions hard to explain • Privacy, approval and clinical testing needed

Conclusion

- Preprocessing improves image quality; segmentation finds the area of interest
- Feature extraction and deep learning identify abnormalities and disease patterns
- Decision-support tools present results clearly to healthcare professionals
- Together they give faster and more consistent diagnosis
- AI assists doctors; it does not replace them